



COMPUTER SCIENCE

CS-305L: Parallel & Distributed Computing Lab

SUBMITTED BY

NAME : RIFAQ AJMAL
REG NO : 23MDBCS 435
SECTION : CS-A
SEMESTER : 6th

Date: 19/3/2026

Lab 08

Step 1: In Anaconda Prompt

- **conda activate mpi_env**
mpi will be activated
- **cd Desktop**
you will go into desktop
- **mkdir Lab_8**
- **cd Lab_8**
you'll go into folder by name lab 8 on desktop

```
Anaconda Prompt

(base) C:\Users\Rifaq Ajmal>conda activate mpi_env

(mpi_env) C:\Users\Rifaq Ajmal>cd Desktop

(mpi_env) C:\Users\Rifaq Ajmal\Desktop>mkdir Lab_8
A subdirectory or file Lab_8 already exists.

(mpi_env) C:\Users\Rifaq Ajmal\Desktop>cd Lab_8

(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_8>
```

Step 2: In VS Code

Task 1

Modify Isend/Irecv → send **list of strings**

```
task1_strings.py
1 from mpi4py import MPI
2
3 comm = MPI.COMM_WORLD
4 rank = comm.Get_rank()
5
6 if rank == 0:
7     data = ["Hello", "MPI", "World"]
8     req = comm.isend(data, dest=1, tag=0)
9     req.wait()
10    print("Process 0 sent:", data)
11
12 elif rank == 1:
13     req = comm.irecv(source=0, tag=0)
14     data = req.wait()
15     print("Process 1 received:", data)
```

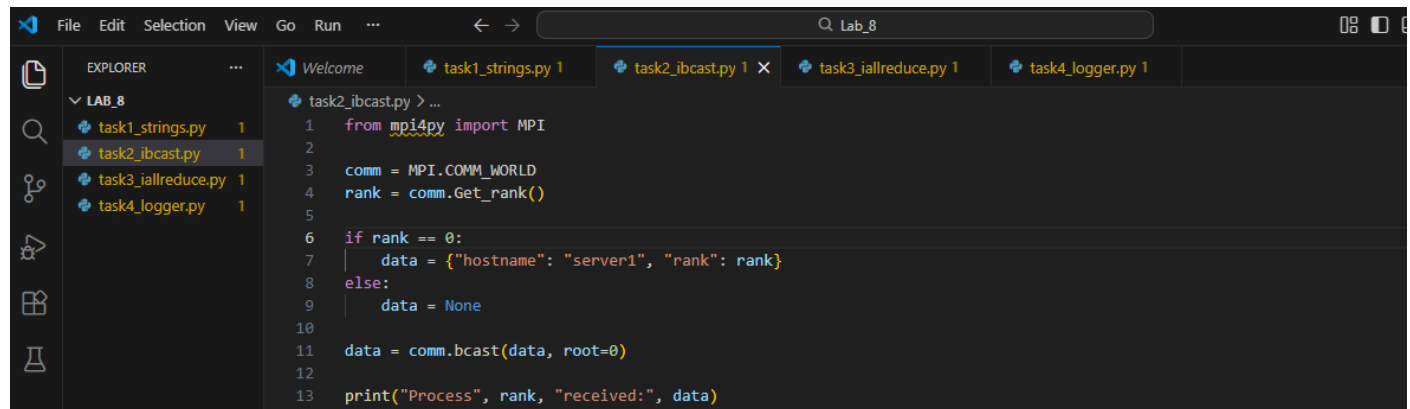
Output/ In Anaconda Prompt

```
(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_8>mpiexec -n 2 python task1_strings.py
Process 0 sent: ['Hello', 'MPI', 'World']
Process 1 received: ['Hello', 'MPI', 'World']

(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_8>
```

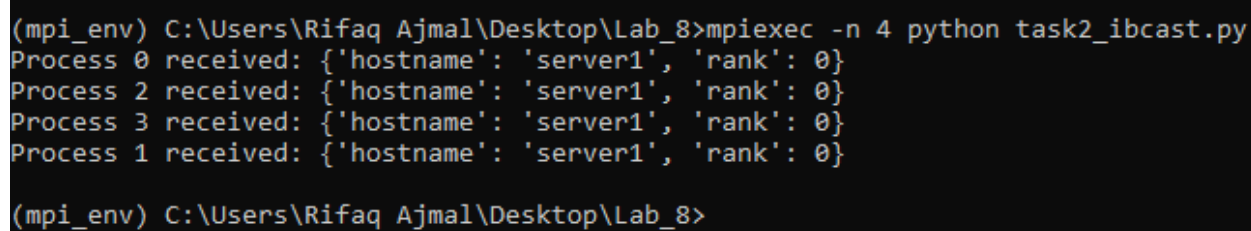
Task 2

Use Ibcast → broadcast dictionary

A screenshot of the Visual Studio Code editor interface. The Explorer pane on the left shows a project named 'LAB_8' with four files: task1_strings.py, task2_ibcast.py, task3_iallreduce.py, and task4_logger.py. The task2_ibcast.py file is selected and its content is displayed in the main editor. The code imports MPI from mpi4py, gets the world communicator, and then uses comm.bcast() to broadcast a dictionary containing hostname and rank from rank 0 to all other processes.

```
task2_ibcast.py > ...
1 from mpi4py import MPI
2
3 comm = MPI.COMM_WORLD
4 rank = comm.Get_rank()
5
6 if rank == 0:
7     data = {"hostname": "server1", "rank": rank}
8 else:
9     data = None
10
11 data = comm.bcast(data, root=0)
12
13 print("Process", rank, "received:", data)
```

Output/ In Anaconda Prompt

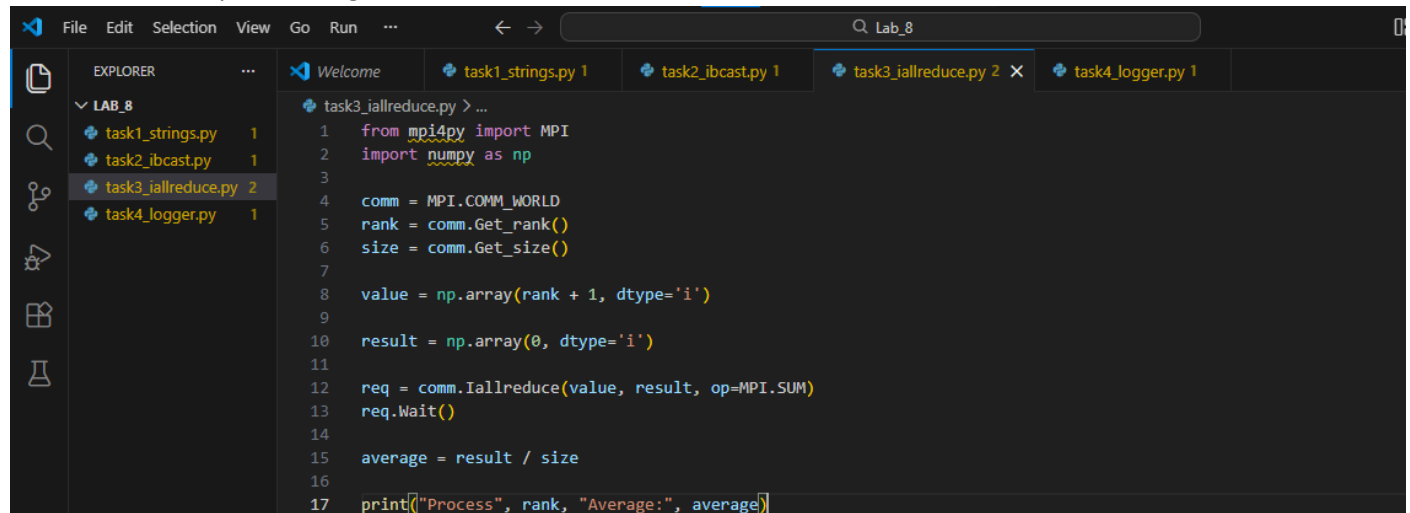
A screenshot of the Anaconda Prompt terminal showing the execution of task2_ibcast.py using mpiexec. Four processes (0, 1, 2, 3) are shown, all receiving the same dictionary: {'hostname': 'server1', 'rank': 0}.

```
(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_8>mpiexec -n 4 python task2_ibcast.py
Process 0 received: {'hostname': 'server1', 'rank': 0}
Process 2 received: {'hostname': 'server1', 'rank': 0}
Process 3 received: {'hostname': 'server1', 'rank': 0}
Process 1 received: {'hostname': 'server1', 'rank': 0}

(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_8>
```

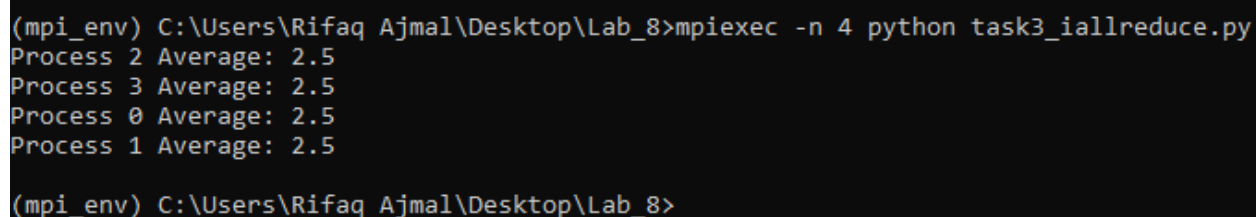
Task 3

Iallreduce → compute average

A screenshot of the Visual Studio Code editor interface. The Explorer pane on the left shows the same project 'LAB_8' with the same four files. The task3_iallreduce.py file is selected and its content is displayed in the main editor. The code imports MPI and numpy, then uses comm.Iallreduce() to compute the average of the rank values across all processes.

```
task3_iallreduce.py > ...
1 from mpi4py import MPI
2 import numpy as np
3
4 comm = MPI.COMM_WORLD
5 rank = comm.Get_rank()
6 size = comm.Get_size()
7
8 value = np.array(rank + 1, dtype='i')
9
10 result = np.array(0, dtype='i')
11
12 req = comm.Iallreduce(value, result, op=MPI.SUM)
13 req.Wait()
14
15 average = result / size
16
17 print("Process", rank, "Average:", average)
```

Output/ In Anaconda Prompt

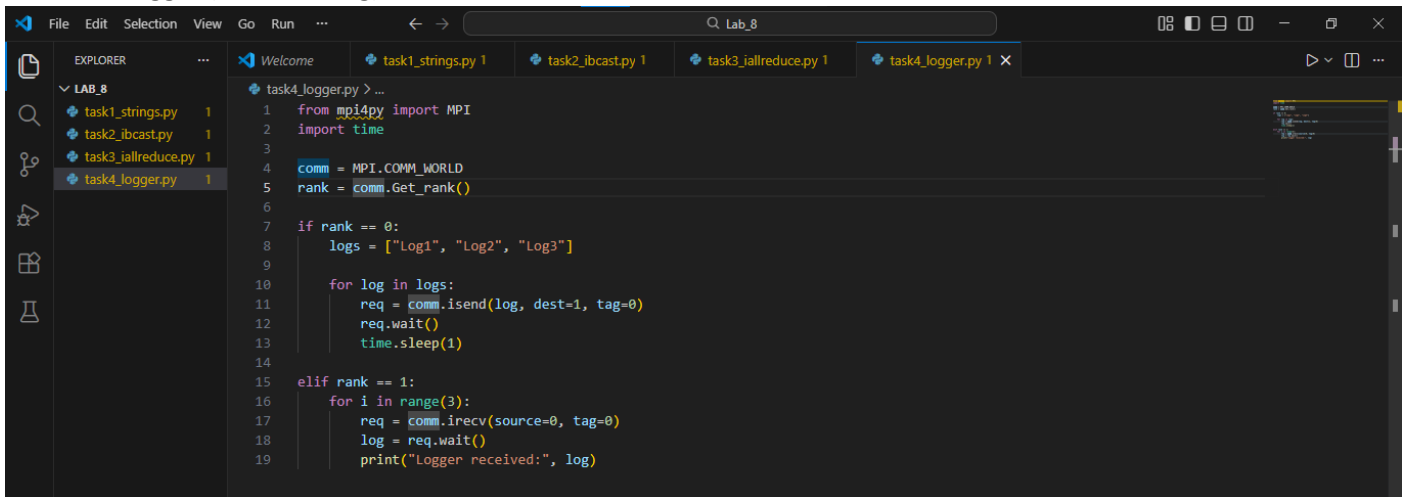
A screenshot of the Anaconda Prompt terminal showing the execution of task3_iallreduce.py using mpiexec. Four processes (0, 1, 2, 3) are shown, all computing an average of 2.5.

```
(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_8>mpiexec -n 4 python task3_iallreduce.py
Process 2 Average: 2.5
Process 3 Average: 2.5
Process 0 Average: 2.5
Process 1 Average: 2.5

(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_8>
```

Task 4

Real-time logger (non-blocking)



```
1 from mpi4py import MPI
2 import time
3
4 comm = MPI.COMM_WORLD
5 rank = comm.Get_rank()
6
7 if rank == 0:
8     logs = ["Log1", "Log2", "Log3"]
9
10    for log in logs:
11        req = comm.isend(log, dest=1, tag=0)
12        req.wait()
13        time.sleep(1)
14
15 elif rank == 1:
16     for i in range(3):
17         req = comm.irecv(source=0, tag=0)
18         log = req.wait()
19         print("Logger received:", log)
```

Output/ In Anaconda Prompt

```
(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_8>mpiexec -n 2 python task4_logger.py
Logger received: Log1
Logger received: Log2
Logger received: Log3

(mpi_env) C:\Users\Rifaq Ajmal\Desktop\Lab_8>_
```